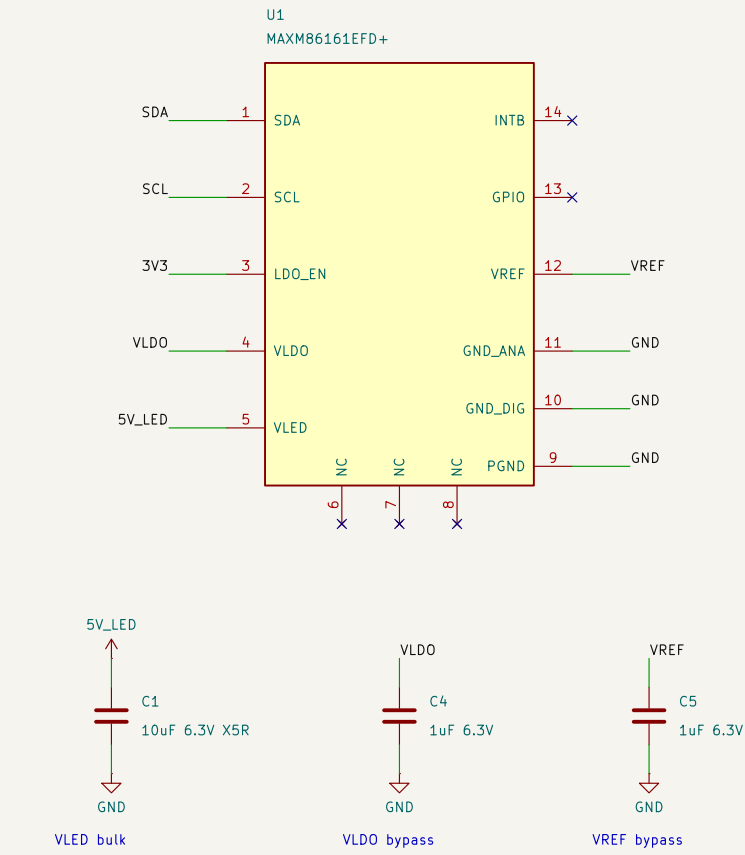


SleepBud | PPG sensor board

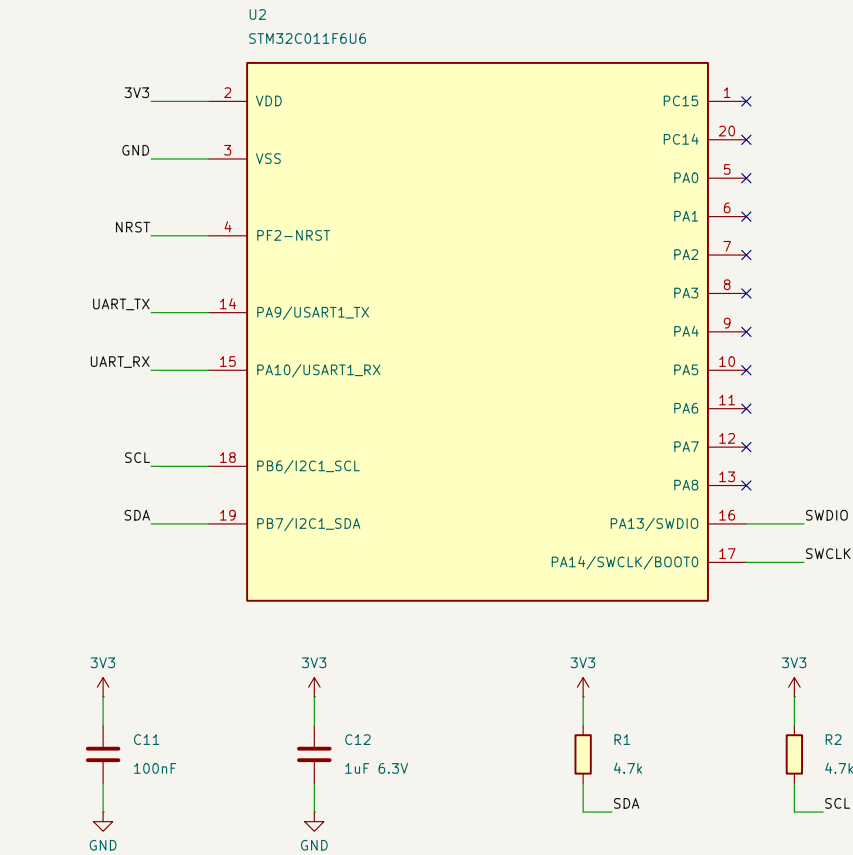
v0.8 / MAXM86161 + STM32C011 / full-duplex UART over RS-422

A3 | FUNCTIONAL SCHEMATIC

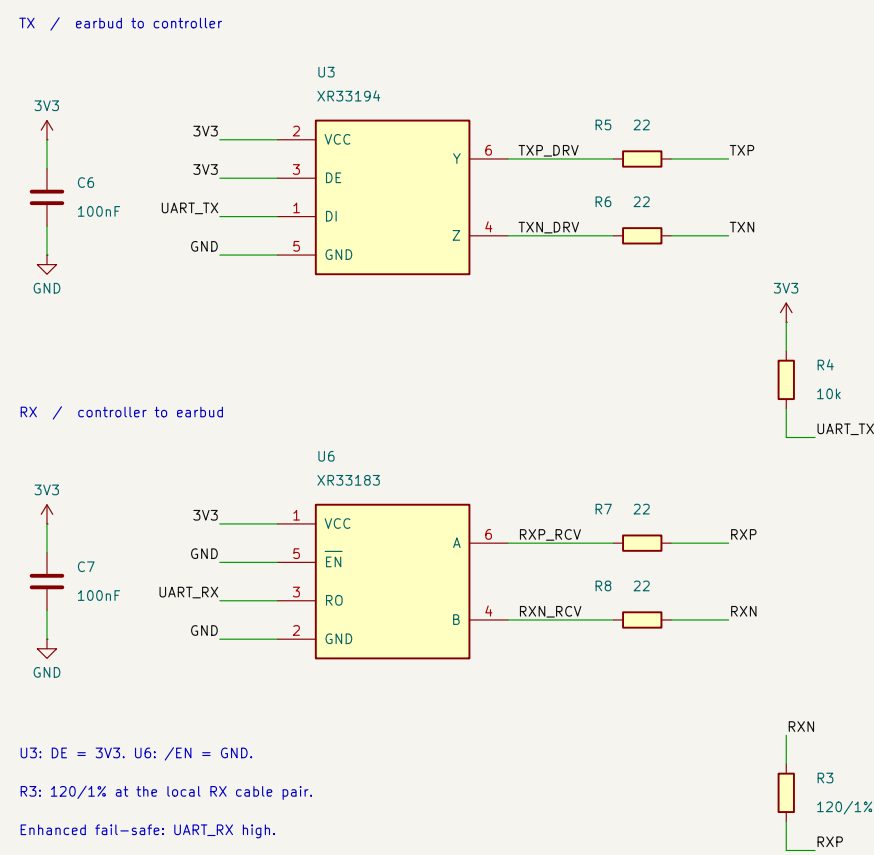
01 PPG SENSOR



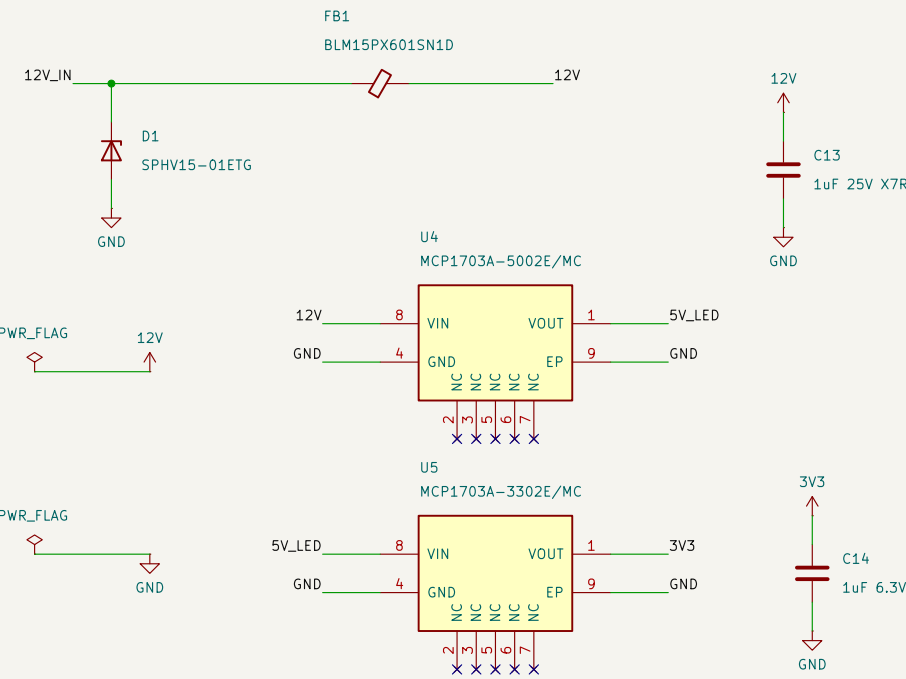
02 MCU + I2C PULL-UPS



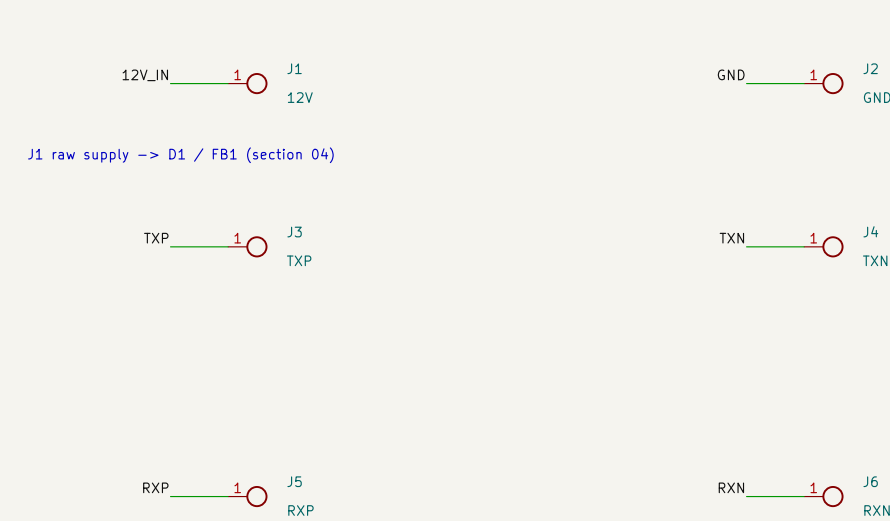
03 UART / RS-422 LINK



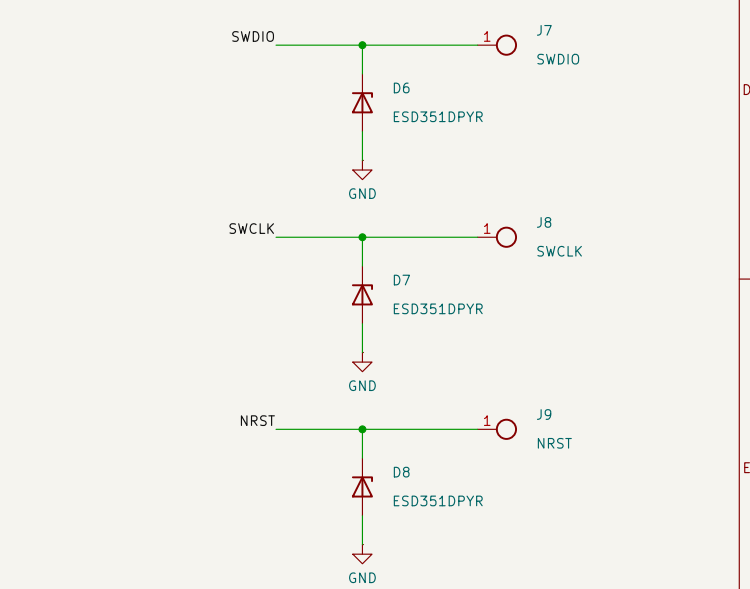
04 INPUT PROTECTION + POWER



05 RS-422 CABLE PADS



06 SWD TEST PADS + ESD



C1: 5V_LED / U4 output bulk capacitor (section 01). D1 clamps raw 12V_IN; FB1 feeds filtered 12V.

Cable: 12V, GND, TXP, TXN, RXP, RXN. Piezo coax bypasses this PCB. Matching net labels are connected.

RS-422 bus TVS omitted; 22 ohm series retained. D1 protects supply, D6-D8 protect SWD. IEC immunity requires physical validation.

PPG sensor board for in-ear sleep device

SleepBud

Sheet: /
File: ppg_pcb_v08.kicad_sch

Title: SleepBud PPG PCB v0.8

Size: A3

Date:

Rev: 0.8

KiCad E.D.A. 10.0.6

Id: 1/1